RTU TRAINING – Q&A

1. What is Live Zero and Dead Zero in Analog Signals?

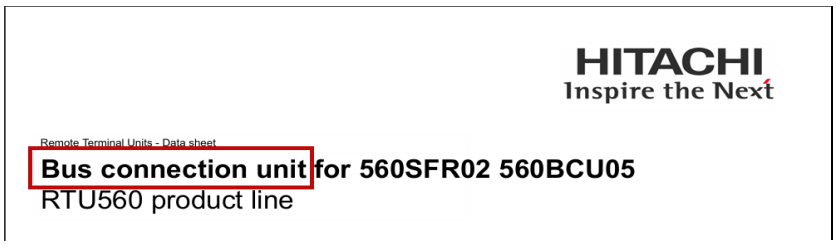
a. In old system(unipolar):
   i. 0mA is minimum value for Analog signals.
   ii. Problem is -> 0mA mean (wire broken, sensor failure, power failure, real zero measurement) -> system cannot differentiate.
   iii. **Dead Zero System ->** 0mA means **zero value** and **wire broken.**
   iv. so the system cannot know if the measurement is valid or not.

b. In new system(live zero – 4 to 20mA):
   i. Minimum signal is shifted to **4mA**.
   ii. In this system (0mA → fault, 4mA → minimum measurement, 20mA → maximum measurement)
   iii. Now the system can detect problems.

c. **Dead Zero →** Zero value is dead → we don't know zero is alive or not.

d. **Live Zero →** Zero value is alive → we know zero value is alive.

2. BCU:



3. Threshold value:

a. RTU continuously receives measurement values from IED (relay, BCU, MFM..)
b. But it can't transmits all value to SCADA, it will causes so much traffic in network
c. Solution is → **Threshold Supervision.**
d. **Threshold Supervision →** when it enables in RTU → RTU only transmits value to SCADA when changes exceeds the **Threshold value.**
   i. **Absolute Threshold:**
      1. Threshold = 2 kV
      2. Last transmitted voltage = 110 kV
      3. If the measured voltage becomes 112 kV, the difference is 2 kV, so the RTU transmits the new value.
   ii. **Absolute(Relative):**
      1. Voltage = 100 kV
      2. Threshold = 2 %
      3. 2 % of 100 kV = 2 kV
      4. If the voltage changes by more than 2 kV, the RTU sends the update.
   iii. **Integrated:**
      1. Threshold = 0.5
      2. Incoming values: 50.00, 50.01, 50.20, 50.30, 50.40, 50.50.
      3. Here total change is 0.1+0.1+0.1+0.1+0.1 = 0.5
      4. Although each individual change is small, the accumulated change reaches the threshold, and the RTU sends the updated value.

iv. **Integrated(Relative):**

Threshold = **2%**
Last value sent to SCADA = **100 kV**

Incoming values from relay:

Value	Change from previous	Accumulated %	Sent to SCADA
101 kV	1%	1%	No
102 kV	1%	2%	**Yes**
103 kV	1%	Reset after send	No

Here, each individual change is small, but the **accumulated percentage change reaches 2%**, so the RTU sends the update to the SCADA system like **MicroSCADA X**.